Software Requirements Specification (SRS)

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1.0	06-10-2026	PES1UG24AM261, PES1UG24AM281, PES1UG24AM289, PES1UG24AM295	Initial SAD document

Approvals

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Operation Theatre Management System (OTMS). It defines the functional requirements, non-functional requirements, external interfaces, security requirements, and verification criteria for managing operation theatres, surgeries, medical staff, equipment, schedules, emergency cases, and resource utilization.

The document serves as a reference for developers, QA engineers, hospital personnel, instructors, and project evaluators.

1.2 Scope

The OTMS manages the scheduling and operational coordination of hospital operation theatres.

The system covers:

- Surgery creation and management
- Operation theatre scheduling
- Doctor and surgeon availability
- Nurse availability
- Surgical equipment availability
- Scheduling conflict detection
- Emergency surgery prioritization
- Rescheduling of affected surgeries
- Schedule and conflict notifications
- Resource allocation
- Theatre and resource utilization reporting
- User authentication and role-based access control
- Audit and monitoring

The system does not implement the internal clinical treatment process or perform medical diagnosis. Integration with external hospital information systems is limited to defined APIs and required data exchange.

1.3 Audience

The intended audience includes:

- Developers
- QA Engineers
- OT Coordinators
- Doctors / Surgeons
- Nurses
- OT Technicians
- Hospital Administrators
- Hospital Management
- System Integrators
- Security Auditors
- Assessment Evaluators

1.4 Definitions

Term	Definition
OTMS	Operation Theatre Management System
OT	Operation Theatre
UI	User Interface
API	Application Programming Interface
RBAC	Role-Based Access Control
CRUD	Create, Read, Update, Delete
ADR	Architecture Decision Record
TLS	Transport Layer Security
SRS	Software Requirements Specification
STP	Software Test Plan
RTM	Requirements Traceability Matrix

2. Overall Description

2.1 Product Perspective

The OTMS is a hospital management application that coordinates operation theatre activities and resources.

The system provides a centralized platform through which authorized hospital users can manage surgeries, operation theatres, medical staff, equipment, schedules, emergency cases, and operational reports.

The system consists of a web-based user interface, application services, scheduling and conflict-detection logic, resource management, notification services, and a relational database.

It may also communicate with external hospital information systems for patient and staff information.

2.2 Major Product Functions

The major functions of OTMS are:

- User authentication and role-based authorization
- Surgery record management
- Operation theatre management
- Surgery scheduling
- Theatre availability checking
- Doctor availability checking
- Nurse availability checking
- Equipment availability checking
- Scheduling conflict detection
- Emergency surgery prioritization
- Rescheduling affected surgeries
- Resource allocation
- Schedule and conflict notifications
- Equipment status and maintenance management
- Theatre and resource utilization reporting
- Audit logging and system monitoring

2.3 User Roles and Characteristics

Role	Responsibilities
OT Coordinator	Responsible for managing operation theatre schedules, assigning surgeries to theatre slots, identifying conflicts, and coordinating emergency cases.

Role	Responsibilities
Doctor / Surgeon	Views assigned surgeries, provides availability information, and receives schedule and conflict notifications.
Nurse	Views assigned surgeries and schedules and receives relevant schedule updates.
OT Technician	Manages equipment availability and maintenance status.
Hospital Administrator	Manages users, roles, theatres, equipment, and administrative information.
Hospital Management	Views theatre utilization, resource utilization, and operational reports.

2.4 Operating Environment

The OTMS shall operate in a hospital network environment.

Software Environment

- Web-based user interface
- Java / Spring Boot backend
- REST APIs
- MySQL or PostgreSQL relational database
- Modern web browsers
- Secure hospital network

Hardware Environment

The system may be accessed through:

- Hospital desktop computers
- Laptops
- Tablets were supported
- Hospital network infrastructure
- Database/server infrastructure

2.5 Constraints

The system is subject to the following constraints:

- Access to sensitive patient and surgery information shall be restricted to authorized users.
- Scheduling shall prevent double-booking of operation theatres.
- Scheduling shall account for doctor, nurse, and equipment availability.
- Emergency cases shall receive higher scheduling priority.
- User access shall be controlled through RBAC.
- Communication between system components shall use secure communication protocols.
- The system depends on accurate availability information entered by authorized hospital personnel.
- External hospital system integration shall use defined APIs.

3. External Interface Requirements

3.1 User Interfaces

The primary interface shall be a web-based OTMS dashboard.

The interface shall provide role-specific functionality.

Interface	Functionality
OT Coordinator Interface	Daily and weekly theatre schedules Surgery scheduling form Conflict warnings Emergency surgery controls Resource availability Rescheduling options
Doctor Interface	Assigned surgery schedule Surgery details Availability management Schedule notifications
Nurse Interface	Assigned surgery list Duty schedule Schedule-change notifications
Technician Interface	Equipment list Equipment availability Maintenance status
Administrator Interface	User management Role management Theatre management Equipment management Audit information
Management Interface	Theatre utilization reports Resource utilization reports Operational statistics

3.2 Hardware Interfaces

The OTMS does not directly control surgical equipment.

The system may interact indirectly with hospital infrastructure and equipment information systems through software interfaces.

Relevant hardware includes:

- Hospital computers and workstations
- Network infrastructure
- Server infrastructure
- Database servers
- Optional notification devices

Equipment availability is maintained through the OTMS interface or integrated hospital systems.

3.3 Software Interfaces

Hospital Information System API

Used for retrieving or exchanging:

- Patient information
- Staff information
- Hospital master data

OTMS REST API

Provides communication between the user interface and OTMS application services.

Example services include:

- Surgery Management API
- Scheduling API
- Resource Availability API
- Conflict Detection API
- Emergency Scheduling API
- Notification API
- Reporting API

Notification Gateway

The system may communicate with an external email/SMS service for notifications.

3.4 Communications

- Network communication shall use TLS 1.2 or higher.
- API communication shall use authenticated requests.
- The system shall validate incoming API requests.
- Failed network requests shall return standardized errors.
- Appropriate retry mechanisms may be used for temporary communication failures.
- Sensitive information shall not be transmitted through unsecured communication channels.

4. System Features (Detailed)

Each requirement includes an acceptance criterion and reference test case.

Requirement IDs follow the format OTMS-F-###.

4.1 Authentication and Authorization

Description

The system shall authenticate users and restrict system functionality according to their assigned roles.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Comments / Dependencies
OTMS-F-001	The system shall authenticate users before providing access to protected OTMS functions.	Functional	High	Security / Administrator	Valid credentials result in successful login. Test: TC-Auth-01	Authentication service required
OTMS-F-002	The system shall enforce role-based access according to the user's assigned role.	Functional	High	Administrator / Security	Users can access only authorized functions. Test: TC-Auth-02	RBAC required
OTMS-F-003	The system shall record successful and failed authentication events in the audit log.	Functional	High	Security	Authentication events are recorded with timestamps. Test: TC-Auth-03	Audit service required

4.2 Surgery Management

Description

The system shall allow authorized users to create, view, update, and manage surgery records.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Comments / Dependencies
OTMS-F-004	The system shall allow authorized users to create a surgery record containing the required surgery details.	Functional	High	OT Coordinator	A valid surgery record is successfully created. Test: TC-SURG-01	Database required
OTMS-F-005	The system shall allow authorized users to update surgery details before the surgery is completed.	Functional	High	OT Coordinator / Doctor	Updated information is saved and displayed correctly. Test: TC-SURG-02	Authorization required
OTMS-F-006	The system shall allow authorized users to view surgery details and status.	Functional	High	Doctor / Nurse / Coordinator	Authorized users can retrieve permitted surgery information. Test: TC-SURG-03	RBAC required

4.3 Theatre Scheduling

Description

The system shall allocate available operation theatre slots while considering surgery duration and resource availability.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Comments / Dependencies
OTMS-F-007	The system shall allow authorized users to schedule a surgery in an available operation theatre slot.	Functional	High	OT Coordinator	Surgery is assigned to a valid available slot. Test: TC-SCH-01	Scheduling service
OTMS-F-008	The system shall check operation theatre availability before confirming a surgery schedule.	Functional	High	OT Coordinator	A theatre cannot be assigned when already occupied. Test: TC-SCH-02	Theatre schedule required
OTMS-F-009	The system shall prevent two surgeries from being scheduled in the same theatre at overlapping times.	Functional	High	Hospital Operations	Conflicting schedule is rejected. Test: TC-SCH-03	Conflict detection required

4.4 Staff and Equipment Availability

Description

The system shall verify the availability of doctors, nurses, and required surgical equipment before confirming a surgery.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Comments / Dependencies
OTMS-F-010	The system shall check the assigned doctor's availability before confirming a surgery.	Functional	High	Doctor / Coordinator	Surgery is rejected when the doctor has an overlapping assignment. Test: TC-RES-01	Staff availability required
OTMS-F-011	The system shall check nurse availability before confirming a surgery.	Functional	High	Nurse / Coordinator	Surgery is rejected when required nurses are unavailable. Test: TC-RES-02	Staff availability required
OTMS-F-012	The system shall check the availability of required surgical equipment before confirming a surgery.	Functional	High	OT Technician	Surgery is rejected when required equipment is unavailable. Test: TC-RES-03	Equipment records required

4.5 Conflict Detection

Description

The system shall automatically identify scheduling conflicts involving theatres, doctors, nurses, and equipment.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Comments / Dependencies
OTMS-F-013	The system shall detect conflicts between overlapping surgery schedules and allocated resources.	Functional	High	OT Coordinator	Conflicting resources are identified before schedule confirmation. Test: TC-CON-01	Scheduling + resource services
OTMS-F-014	The system shall display the details of detected scheduling conflicts to authorized users.	Functional	High	OT Coordinator	Conflict details identify the affected resource and time. Test: TC-CON-02	UI required

4.6 Emergency Surgery Scheduling

Description

The system shall support emergency cases by prioritizing emergency surgeries and determining their impact on existing schedules.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Comments / Dependencies
OTMS-F-015	The system shall allow authorized users to mark a surgery as an emergency case.	Functional	High	Doctor / OT Coordinator	Surgery priority is changed to emergency. Test: TC-EMG-01	Emergency service
OTMS-F-016	The system shall prioritize emergency surgeries when identifying suitable theatre slots and resources.	Functional	High	Hospital Operations	Emergency case receives priority over lower-priority cases. Test: TC-EMG-02	Scheduling rules required
OTMS-F-017	The system shall identify existing surgeries affected by emergency scheduling.	Functional	High	OT Coordinator	All affected schedules are identified. Test: TC-EMG-03	Conflict detection required
OTMS-F-018	The system shall allow affected non-emergency surgeries to be rescheduled to suitable alternative slots.	Functional	High	OT Coordinator	Affected surgery receives a valid alternative slot. Test: TC-EMG-04	Scheduling service required

4.7 Notifications

Description

The system shall notify relevant users about schedule changes, conflicts, and emergency cases.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Comments / Dependencies
OTMS-F-019	The system shall notify relevant users when a surgery schedule is created, changed, or cancelled.	Functional	Medium	Doctors / Nurses / Coordinator	Relevant users receive schedule notification. Test: TC-NOT-01	Notification service
OTMS-F-020	The system shall notify affected users when an emergency surgery causes a schedule change.	Functional	High	OT Coordinator / Staff	Affected users receive emergency-related schedule notification. Test: TC-NOT-02	Notification gateway

4.8 Resource and Utilization Management

Description

The system shall maintain resource information and provide operational utilization information.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Comments / Dependencies
OTMS-F-021	The system shall allow authorized users to update theatre and resource availability.	Functional	High	Administrator / Technician	Updated availability is reflected in scheduling. Test: TC-RES-04	RBAC required
OTMS-F-022	The system shall allow authorized users to update equipment maintenance or availability status.	Functional	Medium	OT Technician	Equipment status changes are reflected in resource checks. Test: TC-RES-05	Equipment records
OTMS-F-023	The system shall generate operation theatre utilization reports for authorized management users.	Functional	Medium	Hospital Management	Utilization report correctly reflects recorded schedules. Test: TC-REP-01	Reporting service

Total Functional Requirements: 23

5. Non-Functional Requirements

The NFRs are measurable and linked to test plans.

IDs follow the format OTMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
OTMS-NF-001	The system shall return normal scheduling and resource availability responses within 3 seconds for at least 90% of requests under normal load.	Performance	High	90th percentile response time ≤ 3 seconds. Test: TC-Perf-01
OTMS-NF-002	The system shall provide at least 99.9% monthly availability, excluding scheduled maintenance.	Reliability	High	Monthly uptime ≥ 99.9%. Test: TC-Rel-01
OTMS-NF-003	The system shall protect sensitive patient and operational information from unauthorized access.	Security	High	Security and authorization tests pass. Test: TC-Sec-01
OTMS-NF-004	The system shall maintain timestamped audit records for security-sensitive and scheduling activities.	Auditability	High	Audit records can be retrieved for tested events. Test: TC-Audit-01
OTMS-NF-005	The system shall provide a clear and usable interface for hospital staff performing scheduling and resource management tasks.	Usability	Medium	Usability evaluation meets defined acceptance criteria. Test: TC-UX-01
OTMS-NF-006	The system shall support concurrent access by multiple authorized hospital users without corrupting scheduling data.	Scalability / Reliability	High	Concurrent scheduling test produces consistent results. Test: TC-Conc-01

5.1 Security

5.1.1 Security Objectives

Security Objective 1 — Confidentiality

The system shall protect patient, surgery, staff, and operational information from unauthorized access or disclosure.

Security Objective 2 — Integrity

The system shall prevent unauthorized or invalid modification of surgery schedules, resource allocations, and other critical operational data.

Security Objective 3 — Accountability

The system shall maintain audit records of important authentication, scheduling, emergency, and administrative activities so that actions can be traced to authorized users.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case
OTMS-SR-001	The system shall use TLS 1.2 or higher for network communication involving sensitive OTMS information.	Security	High	Network security verification passes. Test: TC-SEC-01
OTMS-SR-002	The system shall enforce role-based access control (RBAC) for all protected system functions.	Security	High	Unauthorized role-based access attempts are rejected. Test: TC-SEC-02
OTMS-SR-003	The system shall maintain audit logs for authentication, schedule changes, emergency prioritization, and administrative actions.	Security	High	Required security events appear in audit logs. Test: TC-SEC-03
OTMS-SR-004	The system shall prevent unauthorized users from viewing or modifying restricted patient and surgery information.	Security	High	Unauthorized access is denied. Test: TC-SEC-04
OTMS-SR-005	The system shall validate and authorize API requests before allowing access to protected OTMS resources.	Security	High	Invalid or unauthorized API requests are rejected. Test: TC-SEC-05
OTMS-SR-006	The system shall not store authentication credentials or other sensitive information in application logs in plaintext.	Security	High	Log review confirms sensitive credentials are absent. Test: TC-SEC-06

6. Quality Attributes & Acceptance Tests

The primary quality attributes of OTMS are:

Quality Attribute	Description
Security	The system shall protect patient and operational information using authentication, authorization, encryption, and auditing.
Reliability	The system shall remain available and maintain consistent scheduling information during normal operation.
Performance	Scheduling and resource availability requests should respond within the defined performance target.
Usability	The interface shall allow hospital staff to perform scheduling and resource management tasks clearly and efficiently.
Maintainability	The system shall use modular components so that scheduling, resource management, authentication, notifications, and reporting can be maintained independently.

Acceptance Criteria

The system shall be considered acceptable when:

- All high-priority functional requirements have been implemented and verified.
- No critical defects remain open.
- Required security requirements pass testing.
- NFR acceptance criteria are satisfied.
- Emergency scheduling works correctly.
- Scheduling conflicts are detected correctly.
- RTM coverage is complete for planned requirements and test cases.

Acceptance Test Suites

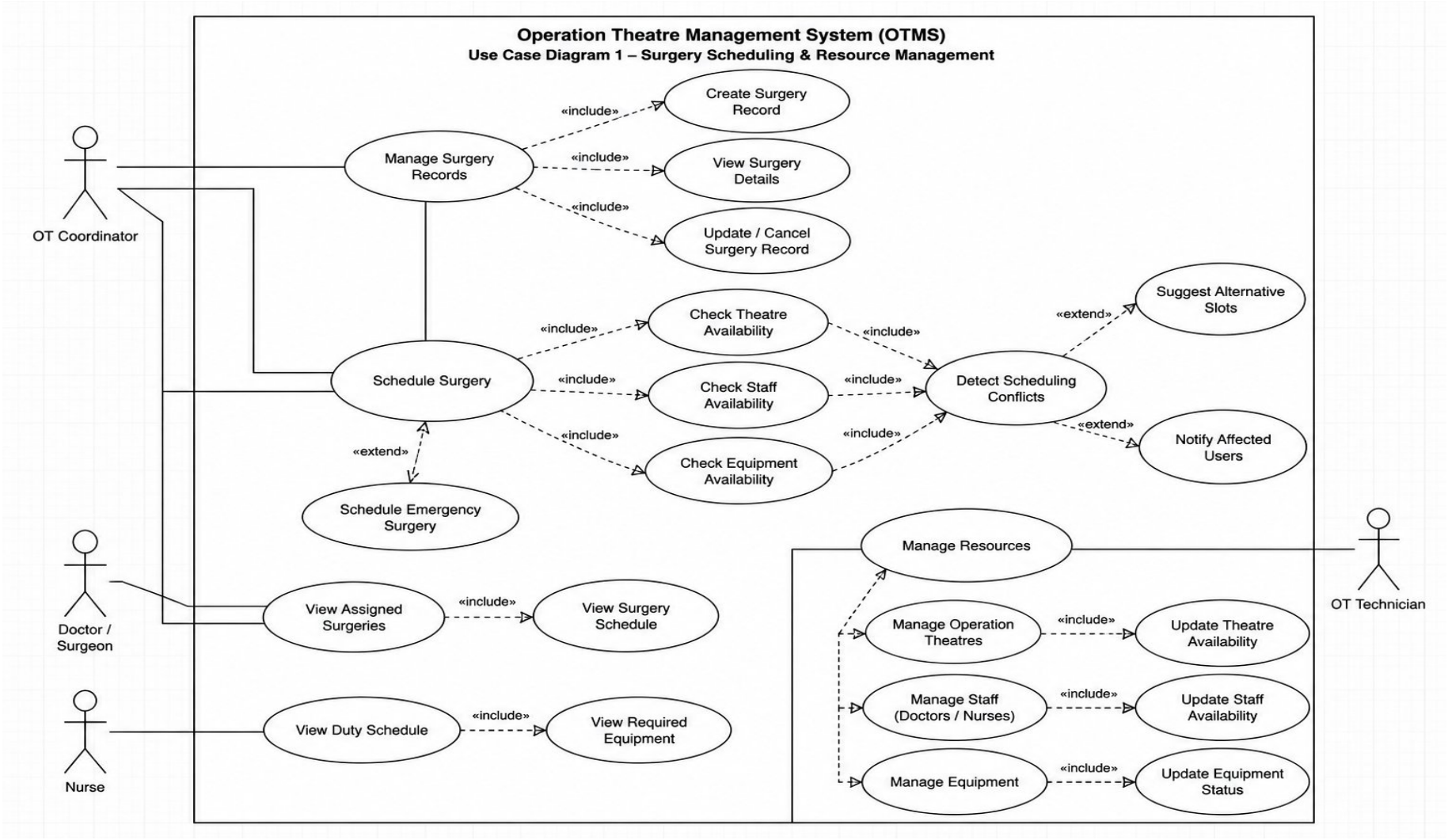
- Authentication testing
- Surgery management testing
- Theatre scheduling testing
- Resource availability testing
- Conflict detection testing
- Emergency scheduling testing
- Notification testing
- Performance testing
- Security testing
- Usability testing
- Reporting testing

7. System Models and Diagrams

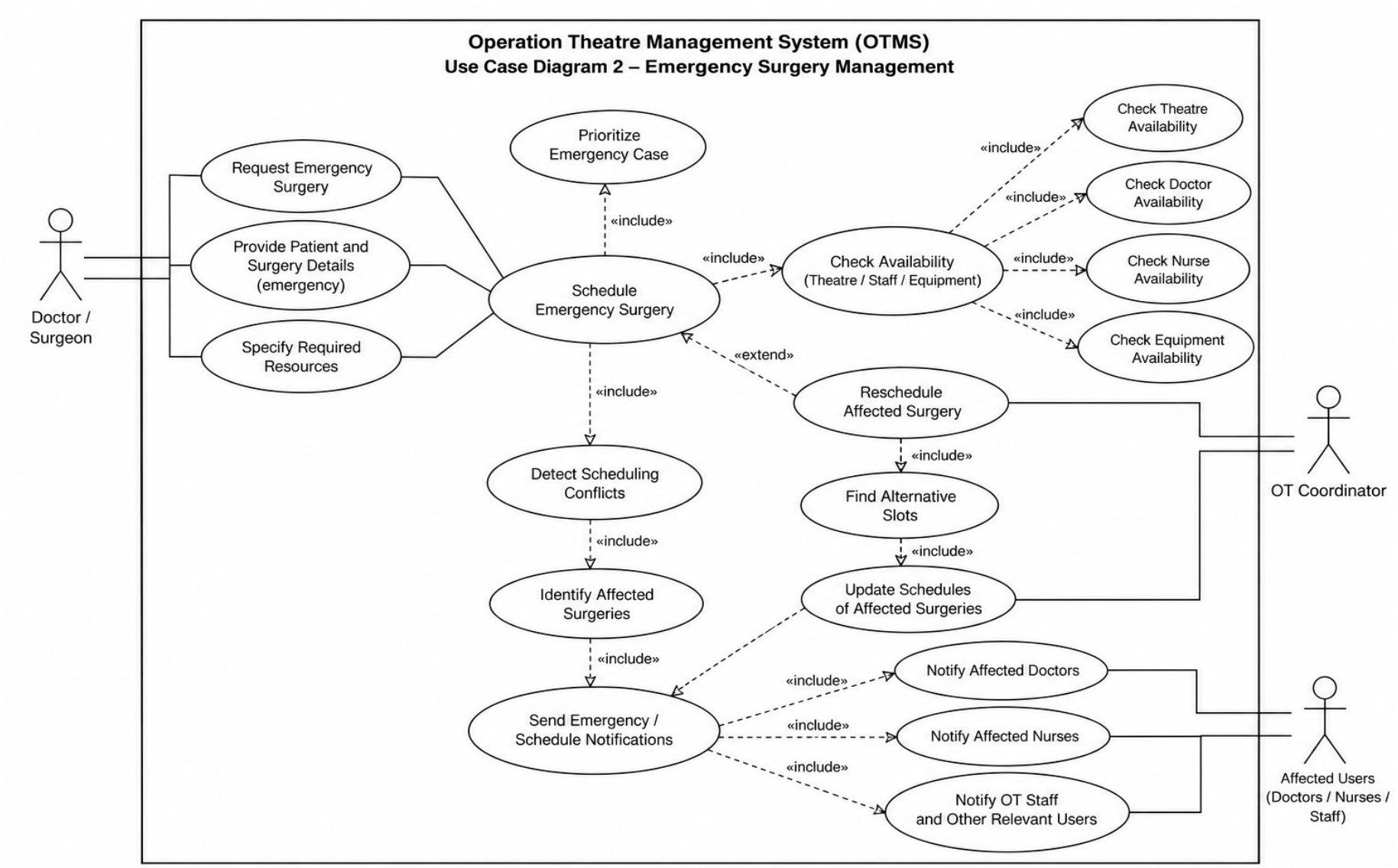
7.1 UML Use-Case Diagrams

At least two UML use-case diagrams shall be created for the OTMS.

Use-Case Diagram 1 — Surgery Scheduling & Resource Management



Use-Case Diagram 2 — Emergency Surgery Management



8. Requirements Traceability Matrix (RTM)

The RTM maps SRS requirements to their architectural components and test cases.

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments
OTMS-F-001	User authentication	4.1 / DS-Auth-01	Authentication & RBAC	TC-Auth-01	N	
OTMS-F-002	Role-based access	4.1 / DS-Auth-02	Authentication & RBAC	TC-Auth-02	N	
OTMS-F-004	Create surgery	4.2 / DS-Surg-01	Surgery Management	TC-SURG-01	N	
OTMS-F-007	Schedule surgery	4.3 / DS-Sched-01	Scheduling Engine	TC-SCH-01	N	
OTMS-F-008	Check theatre availability	4.3 / DS-Sched-02	Resource Management	TC-SCH-02	N	
OTMS-F-009	Prevent theatre conflict	4.3 / DS-Conflict-01	Conflict Detection	TC-SCH-03	N	
OTMS-F-010	Check doctor availability	4.4 / DS-Resource-01	Resource Management	TC-RES-01	N	
OTMS-F-011	Check nurse availability	4.4 / DS-Resource-02	Resource Management	TC-RES-02	N	
OTMS-F-012	Check equipment availability	4.4 / DS-Resource-03	Resource Management	TC-RES-03	N	
OTMS-F-013	Detect scheduling conflicts	4.5 / DS-Conflict-01	Conflict Detection	TC-CON-01	N	
OTMS-F-015	Mark emergency surgery	4.6 / DS-Emergency-01	Emergency Prioritization	TC-EMG-01	N	
OTMS-F-016	Prioritize emergency surgery	4.6 / DS-Emergency-02	Emergency Prioritization	TC-EMG-02	N	
OTMS-F-017	Identify affected surgeries	4.6 / DS-Emergency-03	Conflict Detection	TC-EMG-03	N	
OTMS-F-018	Reschedule affected surgery	4.6 / DS-Emergency-04	Scheduling Engine	TC-EMG-04	N	
OTMS-F-019	Schedule notifications	4.7 / DS-Notify-01	Notification Service	TC-NOT-01	N	
OTMS-F-023	Utilization reports	4.8 / DS-Report-01	Utilization & Reporting	TC-REP-01	N	
OTMS-NF-001	Response time	5 / DS-Perf-01	REST API / Core Services	TC-Perf-01	N	
OTMS-NF-002	Availability	5 / DS-Rel-01	OTMS Application	TC-Rel-01	N	
OTMS-SR-001	TLS 1.2+	5.1 / DS-Sec-01	API Gateway	TC-SEC-01	N	
OTMS-SR-002	RBAC	5.1 / DS-Sec-02	Authentication & RBAC	TC-SEC-02	N	
OTMS-SR-003	Audit logging	5.1 / DS-Sec-03	Audit & Monitoring	TC-SEC-03	N	

Status:

- N = Not yet tested
- P = Passed
- A = Accepted

Final requirement count

Requirement Type	Count	Template Requirement
Functional Requirements	23	≥ 15
Non-Functional Requirements	6	≥ 5
Security Objectives	3	≥ 2
Security Requirements	6	≥ 5